

Analysis of the water balance of representative basins of the mesoregions of Pernambuco based on the influence of SAWP-IAC using SUPER

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ABSTRACT

Climate change is already a reality, evidenced by the continuous rise in air and ocean temperatures and the intensification of extreme events. In this context, hydrological and climatological tools are essential to understand and analyze climate and water dynamics, supporting public policies and the Sustainable Development Goals (SDGs) aimed at mitigating environmental impacts. This study evaluates the water balance of the Brígida, Ipanema, and Sirinhaém river basins for the period 1961–2020, using the SUPER model in association with the variability of the Southwestern Atlantic Warm Pool (SAWP). Temperature anomalies and their influence on precipitation were identified through SAWP variability and the Rainfall Anomaly Index (IAC), while the water balance was assessed for wet and dry periods. Results of evapotranspiration and streamflow simulated by SUPER indicate a strong relationship between SAWP variability and the occurrence of critical events in the basins, affecting water availability. The findings demonstrate that SAWP modulates precipitation, evapotranspiration, and streamflow in Pernambuco, providing valuable insights for integrated water resource management.

Keywords: Southwestern Atlantic Warm Pool, water balance, SWAT.

1. Introduction

The impacts of climate change are already evident. Observations indicate increases in mean air and ocean temperatures, widespread melting of snow and ice, rising mean sea levels, and the expansion of arid regions and desertification hotspots (Corlett, 2014; IPCC, 2023).

The consequences of these changes are identified globally, for example, studies developed by Collert (2014), Guedes and Silva (2020), IPCC (2022), Hartmann et al. (2013), Lacerda et al. (2014), M'Barek et al. (2024), Medeiros et al. (2022), Pandya et al. (2023) and Trenberth (2011), have pointed to a trend of rising global average temperature and changes, such as increases or decreases, in the precipitation regime, especially in semi-arid regions, located in the Northeast region of Brazil (NEB), and their impacts on water resources.

Therefore, trends in climate and water-related changes and extreme events have become guiding principles of the United Nations' 2030 Agenda,

particularly within Sustainable Development Goals (SDGs) 6, 12, and 13, as well as for public policies aimed at reducing environmental degradation, ensuring access to and the sustainable management of water resources, and mitigating climate-related damages and impacts (UN, 2025).

Hydrological and climatological tools are widely used to understand, analyze, and inform decision-making related to climate dynamics and river basin functioning, as well as to support the achievement of the SDGs and the development of public policies aimed at attenuating and mitigating environmental impacts (Costa et al., 2015; Daggupati et al., 2015; Brighenti et al., 2019, M'Barek et al., 2024).

Hydrological modeling is one of the most widely used tools in hydrology and involves the application of different methods to simulate hydrological processes. It can be applied to understand and assess streamflow dynamics and

their changes within a river basin (Silva et al., 2005; Folton et al., 2015). Studies employing

hydrological modeling to characterize hydrological behavior and assess impacts on water resources, such as those by Bajracharya et al. (2024), M'Barek et al. (2024), Souza and Ferreira (2020) and Galvêncio et al. (2007), have reported pronounced changes in river flows, groundwater recharge and surface runoff, as well as increasing conditions of water stress and scarcity associated with changes in precipitation and temperature in the Koshi, El Grou and Brígida Basins.

While climate indices are among the most widely used climatological tools, they enable the identification, visualization, and representation of patterns in atmospheric circulation and temperature variability (Brown and Zhang, 1997; Wang et al., 2007; Kouadio et al., 2012; Hounsou-Gbo et al., 2015). Studies by Cavalcanti et al. (2002), Cintra et al. (2015), Vítório et al. (2025), Vilela et al. (2025), and Oliveira and Medeiros (2021) indicate that sea surface temperature (SST) plays an important role in atmospheric instabilities and moisture convergence, intensifying meteorological systems and precipitation in the western Tropical South Atlantic (TSA), extreme episodes, and excess or deficit of rainfall and water.

The hydrological and climatological tool that best provides observation and analysis resources for the planning and management of water resources for the state of Pernambuco is the Hydrological Response Units System for Pernambuco (SUPER). SUPER is an eco-hydrological model built on the basis of the SWAT (Soil and Water Assessment Tool) hydrological model, which brings together the set of hydrological data and variables from all the Hydrographic Basins of Pernambuco (M'Barek et al., 2024; Galvêncio et al., 2021; Neitsh et al., 2011; Pan et al., 2024; Tiburcio et al., 2023), and the Southwestern Atlantic Warm Pool (SAWP), an index that describes the geophysical process of ocean-atmosphere interaction and the systems involved that occur in the South Tropical Atlantic Ocean (ATS) (Cavalcanti et al., 2002; Cintra et al., 2015; Chidichimo et al., 2023; Domingues et al., 2024).

Given the importance of hydrological and climate modeling for understanding and assisting in assessing environmental damage, this work aims to evaluate the water balance of the Brígida, Ipanema, and Sirinhaém basins using SUPER, from 1961 to 2020, to identify the impacts of extreme events, increases,

and decreases in precipitation. These estimates are made according to a table constructed based on DNOCS (1980); Magalhães et al. (1988); Marengo et al. (2017); Melo (2016); Lacerda et al. (2010); Silva et al. (2020), and the SST phases, according to the classification proposed by Cintra et al. (2022); Medeiros et al. (2022); Trenberth (2011), for the Brígida, Ipanema and Sirinhaém region.

2. Material and methods

Study Area

The study area encompasses the river basins of the Brígida, Ipanema, and Sirinhaém rivers, located respectively in the Sertão, Agreste, and Mata Sul regions of the state of Pernambuco (Figure 1).

The Brígida River basin has an area of 13,745.10 km², while the Ipanema River basin has an area of 7,906.03 km², located in the São Francisco depression, in the Sertão and Agreste regions of Pernambuco. Their predominant climate is semi-arid at low latitudes and altitudes (BSh), with high potential evapotranspiration and scarce precipitation. The Sirinhaém River basin, originating in the Agreste region of Pernambuco, extends entirely within the Zona da Mata Sul and flows into the state's coastline, covering an area of 2,084.18 km².

Study Period

The study period covers January 1961 to December 2020. The data period was adopted according to the standards established by the World Meteorological Organization (WMO), encompassing two climatological normals, which are “average values calculated for a relatively long and uniform period, comprising at least three consecutive decades,” and normal climatological patterns, defined as “averages of climatological data calculated for consecutive 30-year periods” (UFJF, 2025; Domingues et al., 2024).

South Atlantic Warm Pool – SAWP and Rainfall Anomaly Index (RAI)

According to Domingues et al. (2024), the SAWP index refers to the SST anomalies in the region between 15°S and 0° Latitude and 34°W and 30°W Longitude, as shown in Figure 2

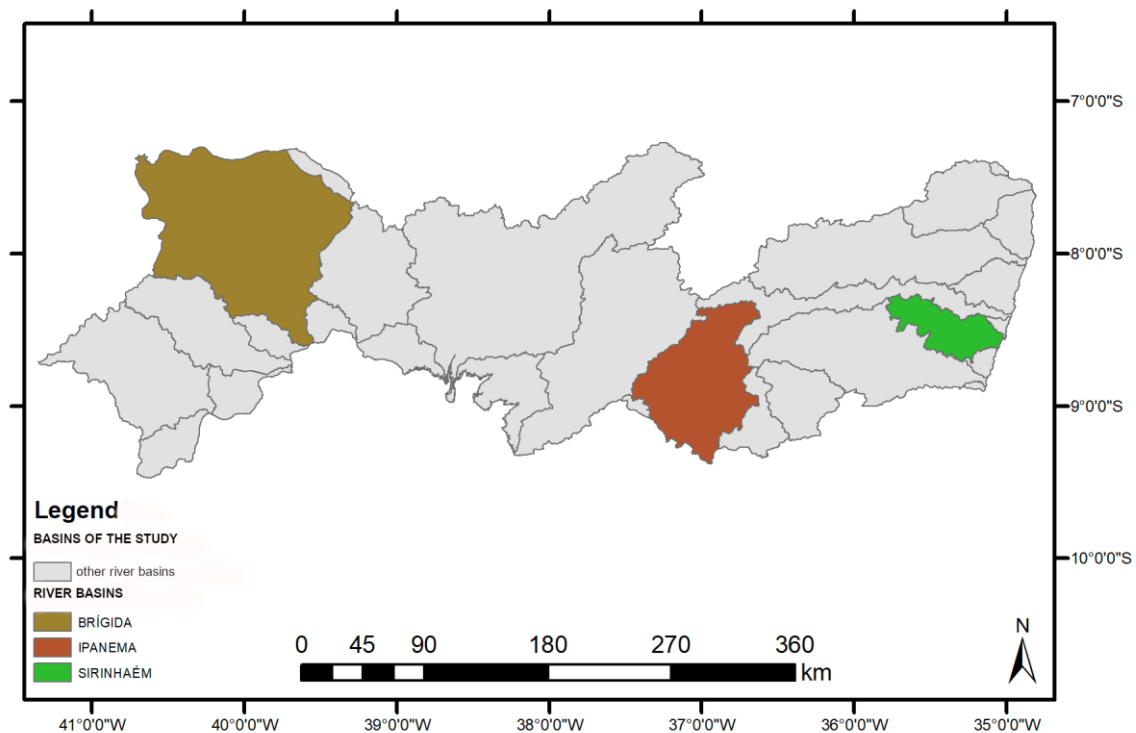


Figure 1 - Map of the Brígida, Ipanema, and Sirinhaém rivers basin.

For the index estimation, the SST trend was removed to analyze the variability without the influence of long-term changes in the Atlantic, as indicated by the IPCC (2022). After trend removal, a spatial average of the area was performed, indicating the periods with positive and negative anomalies in the SAWP. The Rainfall Anomaly Index (RAI) is a method developed by Rooy (1965) and adapted by Freitas (2004) for Northeast Brazil, which was used by Domingues et al. (2024) to analyze the degree of severity, extent, and intensity of dry and rainy years in Northeast Brazil (Figure 3).

ERA5 Database

The data used for the SAWP and IAC analysis come from ERA5, which is a fifth-generation ECMWF global climate atmospheric reanalysis model (www.ecmwf.int/). It is a model produced based on the assimilation of 4D-Var data and model forecasts in CY41R2 from the ECMWF Integrated Prediction System (IFS), with 137 hybrid sigma/pressure (model) levels vertically and the upper level at 0.01 hPa.

Monthly average data of the zonal, meridional, and vertical wind components, specific humidity in both the surface and 250 hPa layers, and single-layer model SST were used.

SUPER - Hydrological Response Unit System of Pernambuco

SUPER is a hydrological simulation model based on SWAT (Soil and Water Assessment Tools - TAMU, 2012). SUPER contains a large part of the hydrographic basins of the state of Pernambuco, with projects already calibrated and validated (Galvêncio *et al.*, 2024; Farias *et al.*, 2023; Tiburcio *et al.*, 2023). SUPER projects simulate various processes in the hydrographic basins with a predefined period, but allowing estimates for future climate and land use and occupation events. This enables the management of the state's water resources to create scenarios for planning actions and public policies aimed at the use and conservation of water.

The SUPER Project used for this study refers to the Brígida, Ipanema and Sirinhaém River Basin (Figure 4) during the simulation period between 1961 and 2020. For the simulation, the available period with

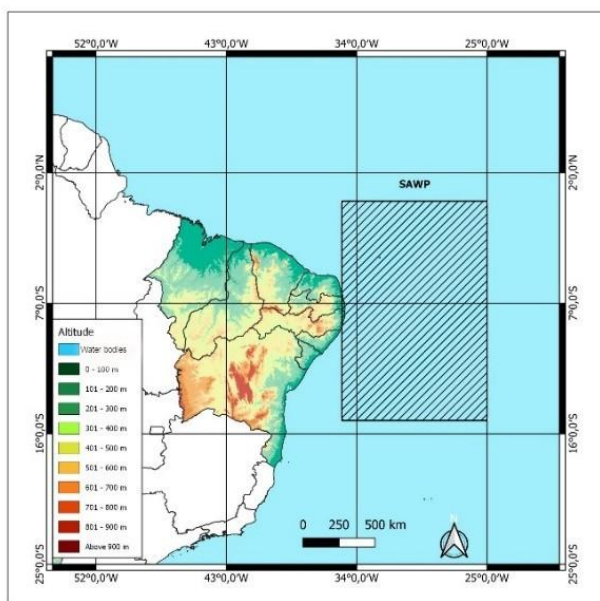


Figure 2 - SAWP Region.

two years of model warm-up was used, with the model estimating data from 1963 to 2020.

For comparison with SAWP, data from August 1982 to December 2020 were used.

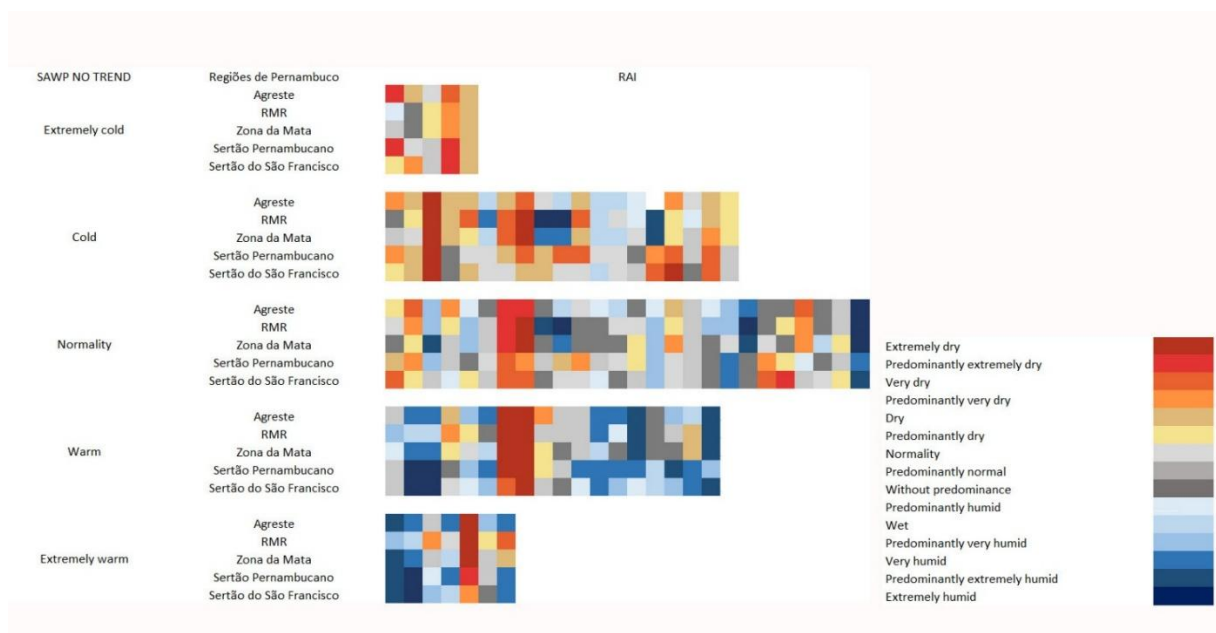


Figure 3 - Clustering matrix of SAWP percentile classifications, along with the mesoregions of Pernambuco and the IAC rainfall regime classifications with normality (Domingues et al. 2024).

These modeling applications were calibrated and validated observed streamflow data following Farias et al. (2023), yielding good to satisfactory performance, with simulated hydrographs adequately representing flow variability. According to Farias et al. (2023), the calibration of the SWAT hydrological model for the Brígida River basin resulted in a Nash-Sutcliffe efficiency (NSE) of 0.61, indicating reasonable model performance. This suggests some limitations of the SUPER model for this basin, requiring further adjustments; nevertheless, the results effectively reflect the vulnerability of the semi-arid region, where hydrological data are scarce and rivers are intermittent. For the Ipanema River basin, the calibration performance was very good (NSE = 0.76), with the model successfully representing mean streamflow conditions and drought periods, although some deficiencies were observed in the simulation of extreme events. In the Sirinhaém River basin, the model achieved good performance (NSE = 0.62), showing minor discrepancies in flood simulations; however, these were substantially smaller than those observed in the semi-arid basins. Overall, the model demonstrated a good fit and produced reliable results for this basin.

3. Results and discussion

The average rainfall across the basins (Figure 5) reveals marked differences among the results. The highest rainfall amounts were observed in the Sirinhaém basin during the austral autumn and winter

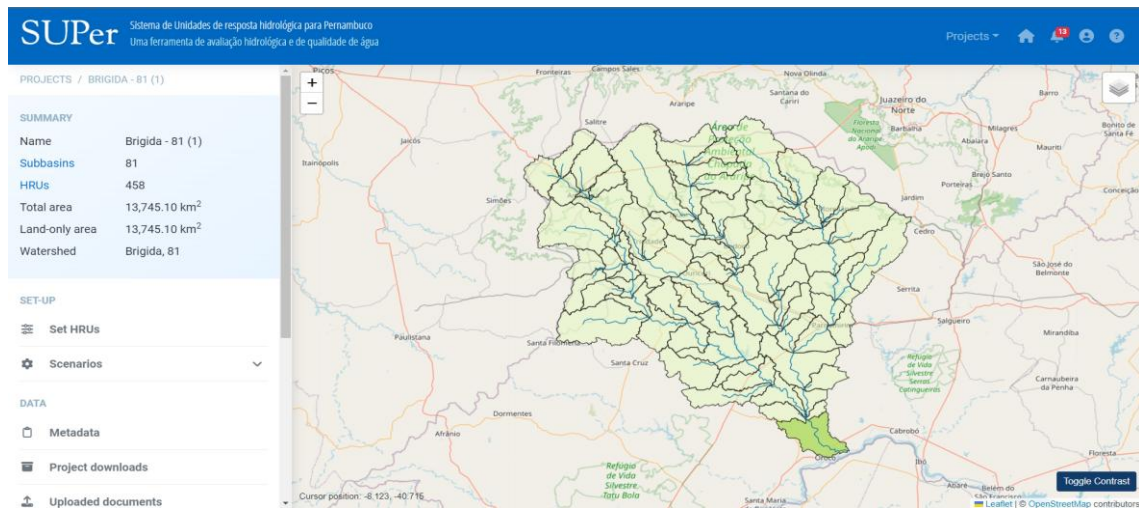
(April to July), which are the rainiest seasons in the Recife Metropolitan Region (RMR) and the Mata Sul region, with monthly totals exceeding 250 mm. In contrast, the lowest rainfall values occurred in the Brígida basin from July to September, spanning the austral winter and spring, which are the driest seasons in the Pernambuco Agreste and Sertão regions and the São Francisco basin, with monthly totals close to 2 mm.

The Ipanema basin exhibited contrasting behavior, with average rainfall values showing no significant increases, in contrast to the high precipitation peaks observed in the Sirinhaém basin and the low precipitation values recorded in the Brígida basin. The lowest average rainfall in the Ipanema basin occurred between March and July, during the austral autumn and winter, with monthly totals not exceeding 80 mm. In this region, autumn is typically the rainiest season, whereas winter is the driest.

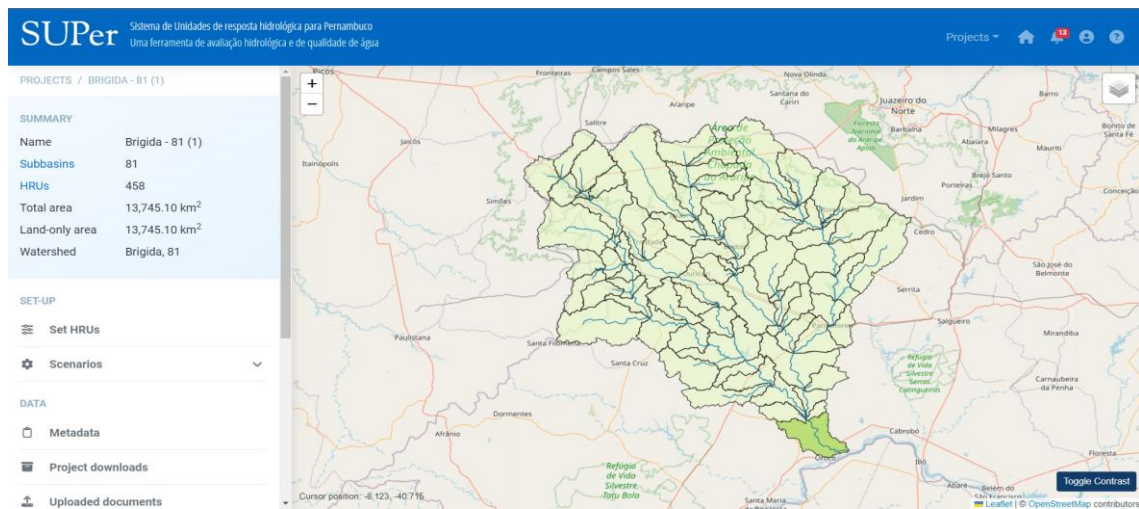
The average precipitation values across the basins reflect regional differences driven by variations in temperature, humidity, wind patterns, and precipitation regimes, which are influenced by ocean-atmosphere interactions, particularly SST (Ayoade, 2002; Cavalcanti *et al.*, 2002; Corlett, 2014). The interaction between the atmosphere and SST involves processes such as the transport of water vapor from the ocean to the land, heat and moisture exchange between the ocean and the atmosphere, and instabilities over warm waters (Cavalcanti et al., 2002; IPCC, 2021; Medeiros et al., 2022).

Warm SSTs contribute to the development of low-pressure centers, favoring moisture convergence, increasing water vapor content, atmospheric instability, and the formation of convective clouds,

A.



B.



C.

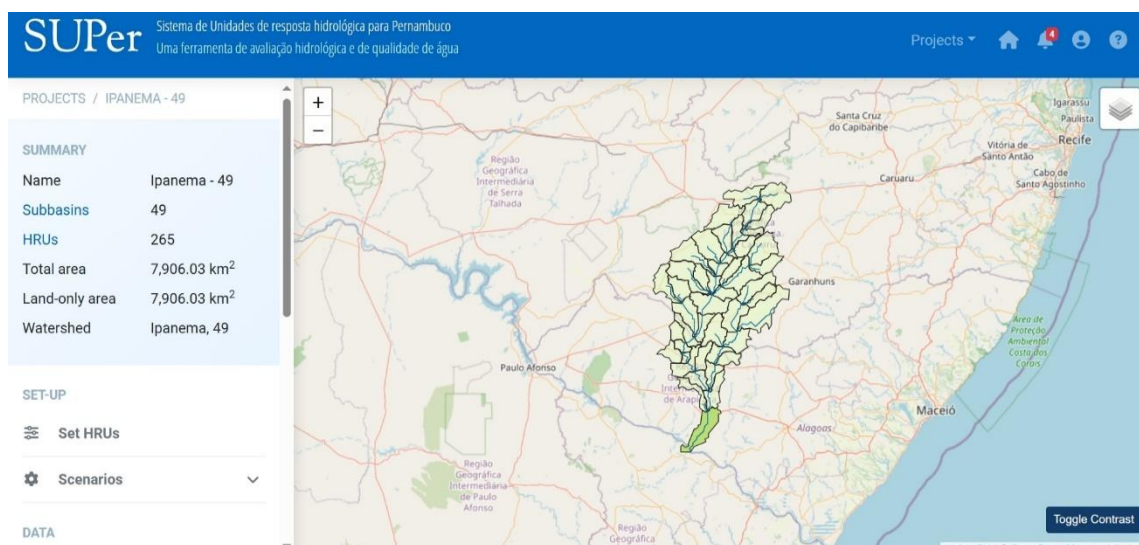


Figure 4 - SUPer Projects for the Hydrographic Basins of the Brígida(A), Ipanema(B) and Sirinhaém(C) Rivers.

Excess or deficit rainfall is closely linked to interactions between oceanic and atmospheric processes. Anomalous SST patterns can influence

intensifying meteorological systems in adjacent continental regions (Kouadio et al., 2012; Hounsou-Gbo et al., 2015).

local precipitation when acting in isolation. However, when such anomalies occur simultaneously across different ocean basins, their impacts extend to large-

scale atmospheric circulation, affecting both the meridional component through the Hadley circulation and the zonal component through the Walker circulation (Cintra et al., 2015; Chidichimo et al., 2023; Hounsou-Gbo et al., 2015; Wang and Enfield, 2003).

Warming in the South Tropical Atlantic (STA) region is associated with anomalous climate conditions, including increased occurrences of positive SST anomalies in the eastern Pacific (El

Niño) and negative anomalies in the South Atlantic. These coupled patterns, as demonstrated by Cintra et al. (2015), Nóbrega and Oliveira (2016), Medeiros et al. (2021), and Vilela et al. (2025), highlight the interconnection between large-scale ocean–atmosphere variability and the occurrence of extreme rainfall events and their associated hydrological impacts

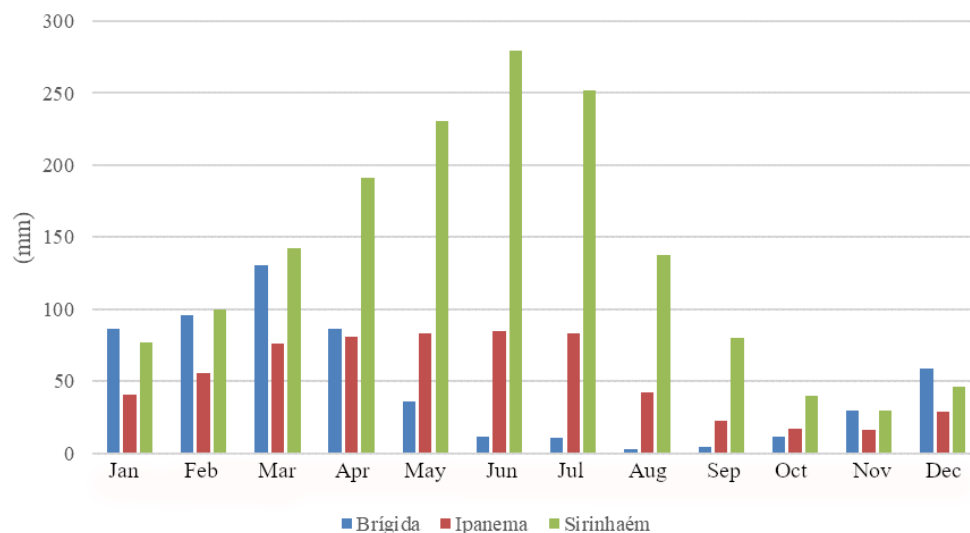


Figure 5 - Average Precipitation of the Basins.

The warming of the South Tropical Atlantic (STA), as documented by Cintra et al. (2015), Nóbrega and Oliveira (2016), Medeiros et al. (2021), and Vilela et al. (2025), is strongly linked to the occurrence of extreme rainfall events and associated hydrological impacts. These influences are evident in extreme precipitation episodes in eastern Northeast Brazil (Silva et al., 2018), severe droughts affecting the region (Marengo et al., 2017), pronounced variability in the Goiana, Capibaribe, Ipojuca, Sirinhaém, Una, and Mundaú river basins, and the intensification of aridity in the Brígida and Garças basins (Silva et al., 2020). Together, these patterns highlight the role of SST-driven ocean–atmosphere interactions in modulating hydroclimatic variability across the South Tropical Atlantic region.

The acceleration in the variation of evapotranspiration, a combined process of evaporation from the soil surface and water bodies and transpiration from plants, pointed out by the studies of Silva et al., 2020, is visualized in Figure 6A, Graph of average actual evapotranspiration. It is observed that the Sirinhaém basin, from April to September (autumn to winter), and the Brígida basin, from December to April (summer to autumn), have the highest evapotranspiration values, while the Ipanema basin has the lowest value in all months.

Potential evapotranspiration (ET_o), which represents the combined processes of soil evaporation and plant transpiration, exhibits a spatial pattern distinct from that of actual evapotranspiration, as shown in Figure 6B and Figure 6A, respectively. The highest values, exceeding 200 mm, were observed in the Brígida basin between May and November, corresponding to the winter and spring seasons, and in the Ipanema basin between December and April, during the summer and autumn seasons. In contrast, the Sirinhaém basin exhibited lower values throughout the year, remaining below 190 mm in all months.

Galvêncio et al. (2024) showed that the water balance is impacted by the decrease in evaporation and transpiration, but there is an increase in potential evapotranspiration in a study in the semi-arid region. Fontes Júnior et al. (2019) in a study conducted in the headwaters of the Ipanema River basin, observed variations in evapotranspiration associated with extreme drought events and its impact on the water balance driven by vegetation dynamics.

Figure 7 shows the flows of the Brígida, Ipanema, and Sirinhaém River basins according to the temperature conditions of the SAWP and its influence on precipitation events that occurred from 1982 to 2020 in Northeast Brazil.

In the Brígida River basin (Figure 7A), the Extremely Hot and Hot periods presented the highest

streamflows recorded in the 1980s, with flows close to 40 m³/s and 32 m³/s, respectively. Over the following decades, the streamflows were reduced, which can be explained by the presence of large dam constructions such as Entremontes and Chapéu. In the 1990s and early 2000s, the Normal SAWP was predominant in

the highest streamflows that occurred in the basin, with streamflows close to 20 m³/s. In the decade after 2010, we observed the relationship between Cold SAWP in Normality, identifying streamflows below 5 m³/s between the years 2012 and 2016.

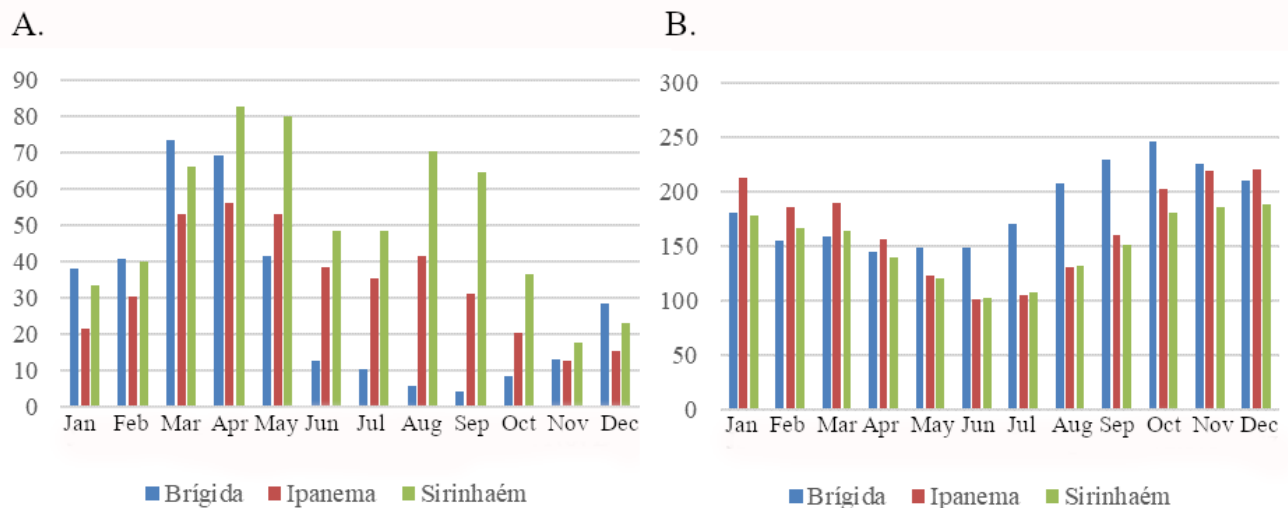


Figure 6 - Graphs of average evapotranspiration (A) and average potential evapotranspiration (B).

Vitório et al. (2025) and Domingues et al. (2024) highlight that the warm and extremely warm phases of the SAWP, which normally occur between March and May, favor evaporation and moisture transport, intensifying the action of the Intertropical Convergence Zone (ITCZ) over the Northeast and impacting the rainfall regime in the Brígida River basin. However, Hounsou-Gbo et al. (2015) highlight that the influence of SAWP can be mitigated by other atmospheric systems, such as low-pressure centers and Easterly Waves, which modify the rainfall distribution in the basin region.

The direct effect of these variations in the rainfall regime on the basin's water resources is observed, highlighting the socio-environmental vulnerabilities of the population in the region (Silva et al., 2023), impacting the water balance and affecting hydrological cycle processes such as surface runoff and water supply for the populations (Brito et al., 2020; Tiburcio et al., 2023).

While the Ipanema River basin (Figure 7B) is influenced by the SAWP (Extremely Warm and Hot Waves) for systems that operate in both the northern and eastern regions of Northeast Brazil, similarly to the Brígida basin, the Extremely Warm and Hot SAWP contribute to the extreme flow rates of the 1980s, approaching 270 m³/s and 209 m³/s respectively.

In the SAWP (Short Water Flow) during Normal, Cold, and Extremely Cold periods in the

1990s, 2000s, and 2010s, flow events were below 150 m³/s, but distributed throughout the rainy periods.

We observed an alternation of flow peaks in the basin from Extremely Hot to Extremely Cold, with a predominance of Normal periods, which may be influenced by systems occurring in the eastern part of northeastern Brazil and the 4-month lag between anomalies (Vitorio et al., 2025). The decrease in rainfall during the Cold and Extremely Cold SAWP events corroborated the drought extremes in the basin (Domingues et al., 2024). Limited results, the model needs additional adjustments; it accurately reflects the vulnerability of the semi-arid region, where hydrological data are scarce and rivers are intermittent.

In the Sirinhaém River basin, the Normal state of the SAWP (Swimming and Water Supply System) shows the highest flows throughout the study period, with estimated flows close to 300 m³/s. Because it is located in a more humid region due to meteorological systems influenced by greater moisture transport from the ocean to the atmosphere, dry periods are not as prolonged as in the interior. The flows presented by SUPer (Superior Water Supply System) demonstrate the basin's vulnerability to low to moderate flooding, regardless of the SAWP state. In a hydrological analysis of the basin, Portela et al. (2023) found that these vulnerabilities are present at the basin's outlets and near the rivers.

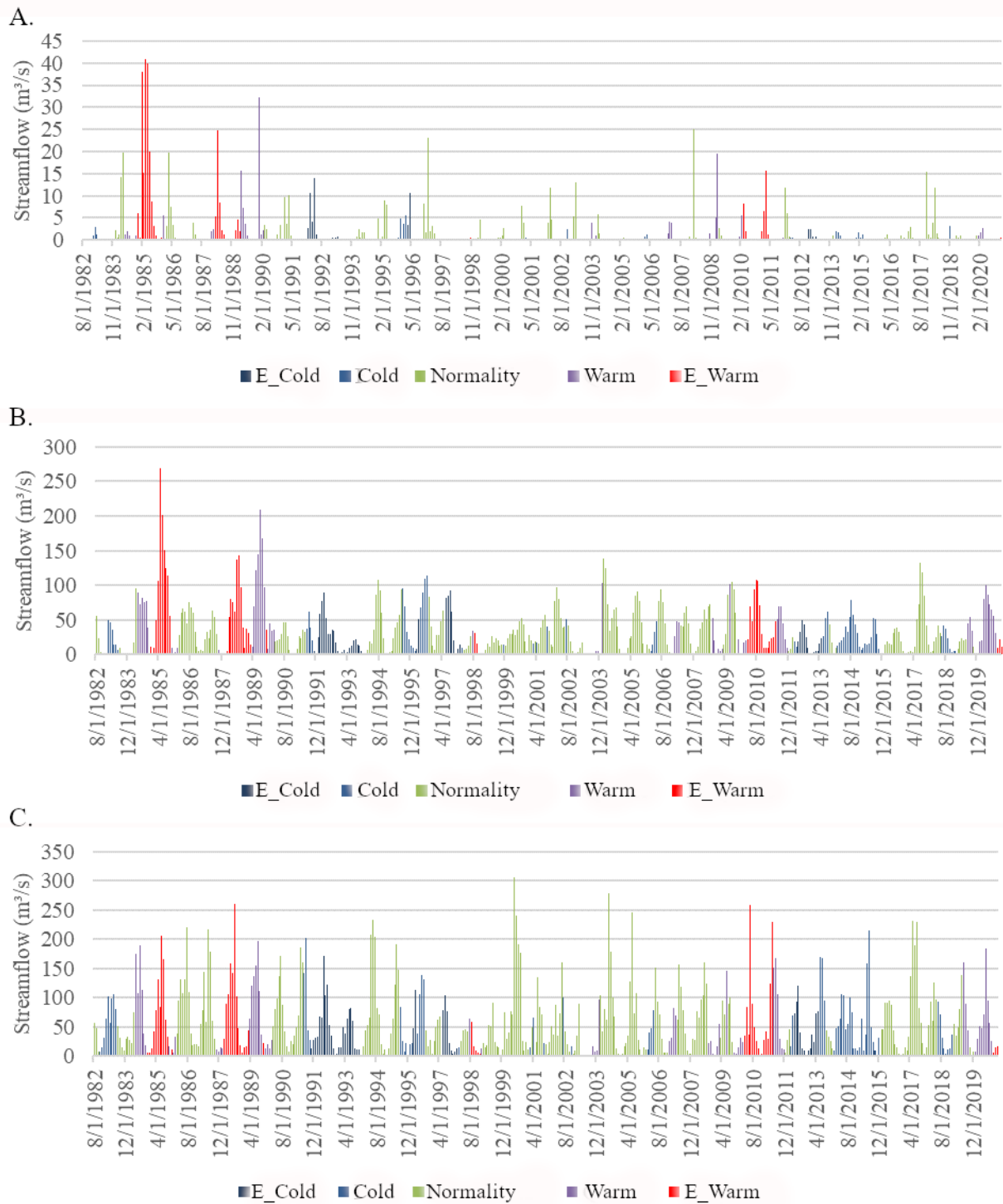


Figure 7. Streamflows of the Brígida (A), Ipanema (B) and Sirinhaém (C) Rivers during the Extremely Cold (E_Frio), Cold, Normal, Hot and Extremely Hot (E_Quente) events of the SAWP.

4. Conclusions

The results of this study indicate significant changes in the water balance of the Brígida, Ipanema, and Sirinhaém basins. These changes reflect the influence of the SAWP on precipitation, evapotranspiration, and streamflow across the three basins. A direct and linear relationship is observed between SAWP temperature and precipitation,

evapotranspiration, and streamflow. Cooler SST conditions are generally associated with water scarcity and reduced evapotranspiration, whereas warmer and near-normal SST conditions tend to favor higher water availability and enhanced evapotranspiration.

Visualizing the influence of SST on the modulation of atmospheric systems and the generation of hydrological impacts in basins highlighted the need to deepen and develop further studies on ocean-atmosphere relationships, using scenario simulations

as a basis, and the effects on water availability in basins, aiming at social and economic damages, as a way to contribute to and support the management of water resources in the state of Pernambuco.

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